

From Niche to Necessity: Small AR Innovations, Big Potential for Impact in Therapy

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College of Computer Studies

CENTER FOR COMPLEXITY & EMERGING TECHNOLOGIES

Images ¾ the works of:
Biener, V., Kalamkar, S., Nouri, N., Ofek, E., Pahud, M., Dudley, J. J., ... & Grubert, J. (2022). Quantifying the effects of working in vr for one week. *IEEE Transactions on Visualization and Computer Graphics*, *28*(11), 3810-3820.
Wagener, N., Reicherts, L., Zargham, N., Bartłomiejczyk, N., Scott, A. E., Wang, K., ... & Niess, J. (2023, April). SelVReflect: A Guided VR Experience Fostering Reflection on Personal Challenges. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1-17).
Honda, S., Sawabe, T., Nishimura, S., Sato, W., Fujimoto, Y., Plopski, A., ... & Kato, H. (2019, September). Evaluation of relationship between stroke pace and speech rate for touch-care robot. In *Proceedings of the 7th International Conference on Human-Agent Interaction* (pp. 283-285).

Augmentations

Physical	Usually requires an auxiliary hardware or attachment
Digital	Usually requires a display and tracking technology

Modalities of Digital Augmentations

Light	How brightly-seen is the augmentation?
Color	What color is seen?
Form	On what form does the augmentation appear?
Move	Does the augmentation move? If so how?

Better yet mix or extend these augmentations

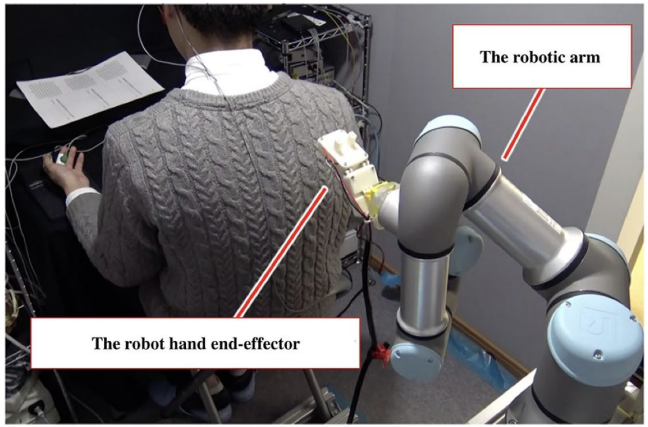
Hear	Beyond seeing can we also hear it?
Haptic	Can we also sense its movement or feel its texture?
Tangible	Can we touch it or can it touch us?

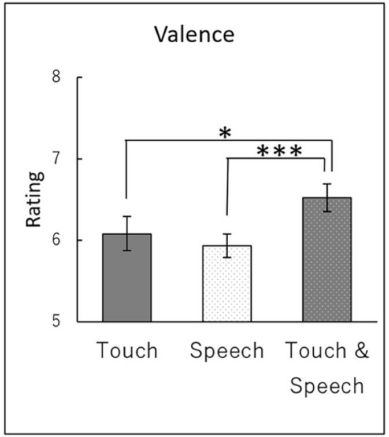
Current solutions in AR may have already addressed some of these issues in rehabilitation and in therapy.

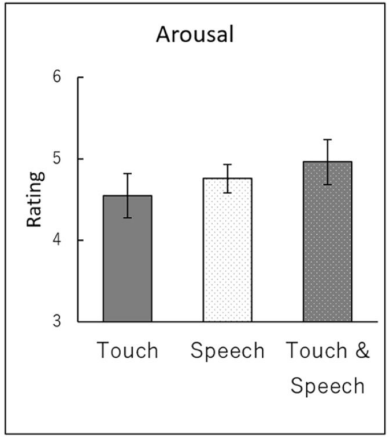
Potential loss of personal touch

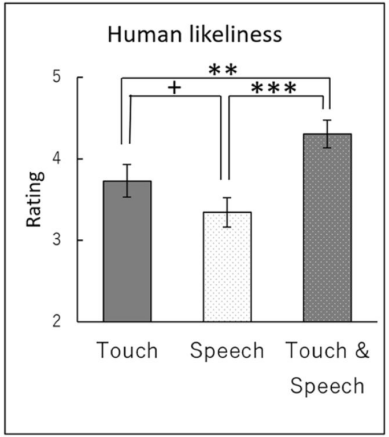
Answer

Robotic touch with speech boosts positive emotions







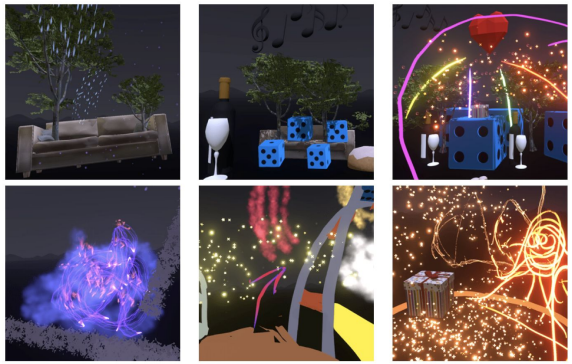


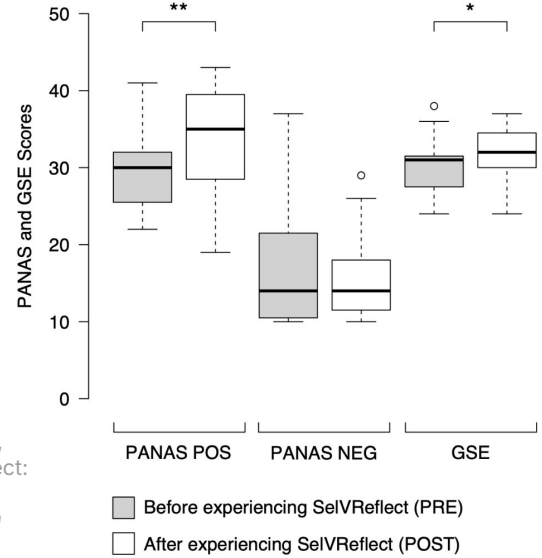
Sawabe, T., Honda, S., Sato, W., Ishikura, T., Kanbara, M., Yoshikawa, S., ... & Kato, H. (2022). Robot touch with speech boosts positive emotions. *Scientific Reports*, *12*(1), 6884.

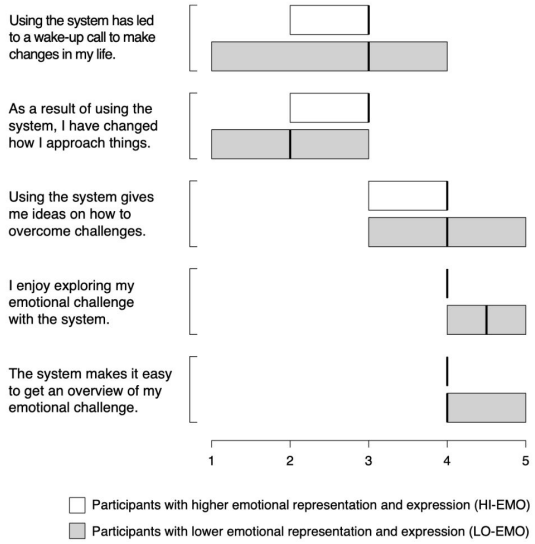
Potential withdrawal from real-world interactions

Answer

Guided VR reflections can support deeper self-reflections





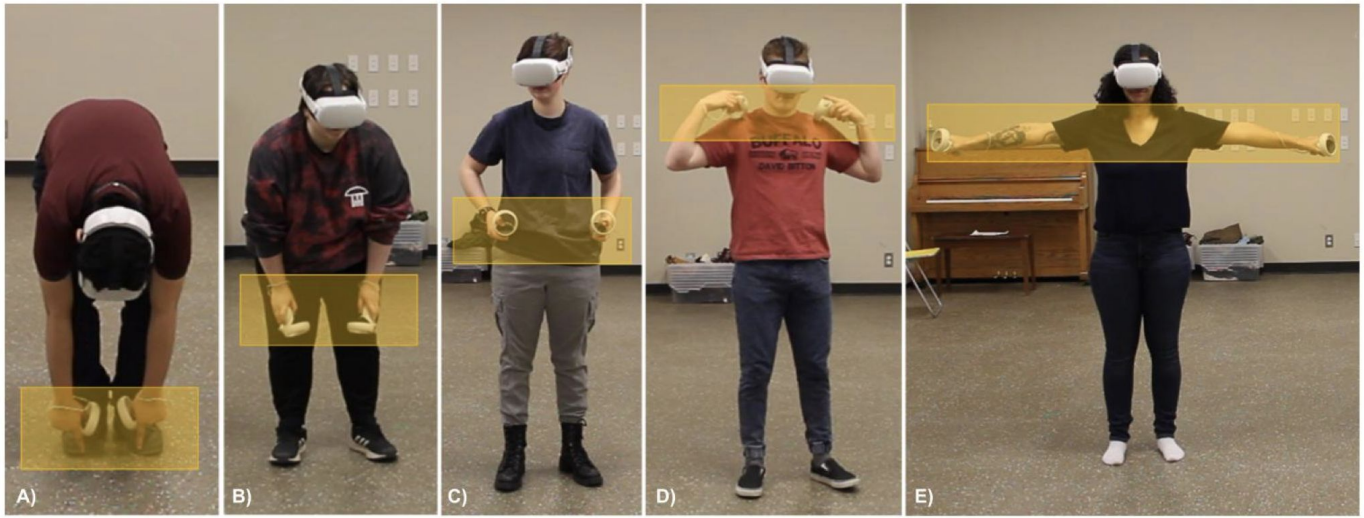


Wagener, N., Reicherts, L., Zargham, N., Bartłomiejczyk, N., Scott, A. E., Wang, K., ... & Niess, J. (2023, April). SelVReflect: A Guided VR Experience Fostering Reflection on Personal Challenges. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1-17).

Risk for Virtual Reality sickness & other symptoms

Answer

Thresholding protocols preVR support continuous use and longevity

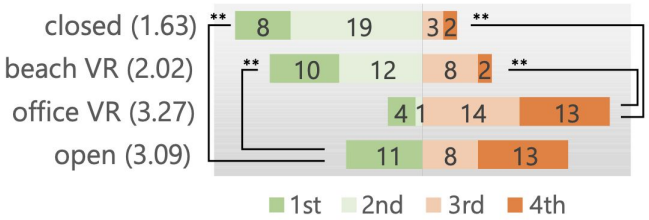


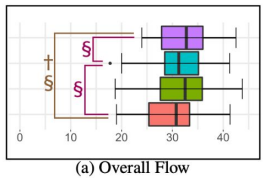
Mckendrick, Z., Somin, L., Finn, P., & Sharlin, E. (2023, June). Virtual Rehearsal Suite: An Environment and Framework for Virtual Performance Practice. In *Proceedings of the 2023 ACM International Conference on Interactive Media Experiences*(pp. 27-39).

Virtual Reality is not reality

Answer

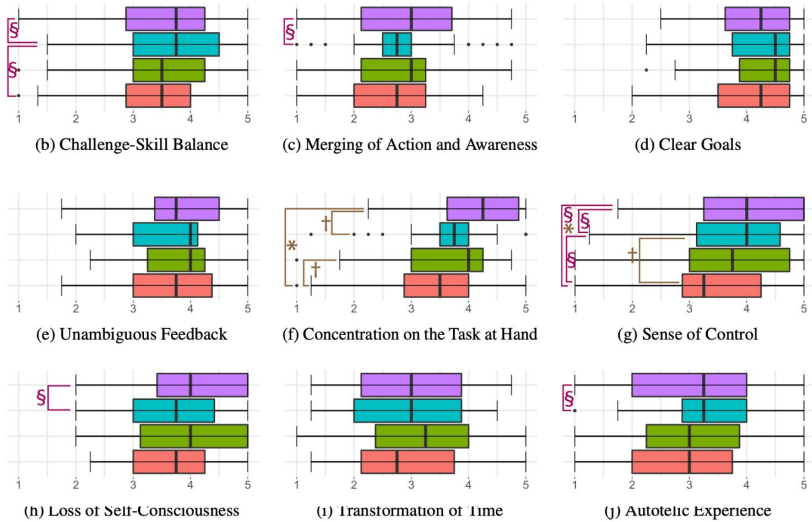
Staying in modified VR workplaces enable flow and keep users focused.





(a) Overall Flow





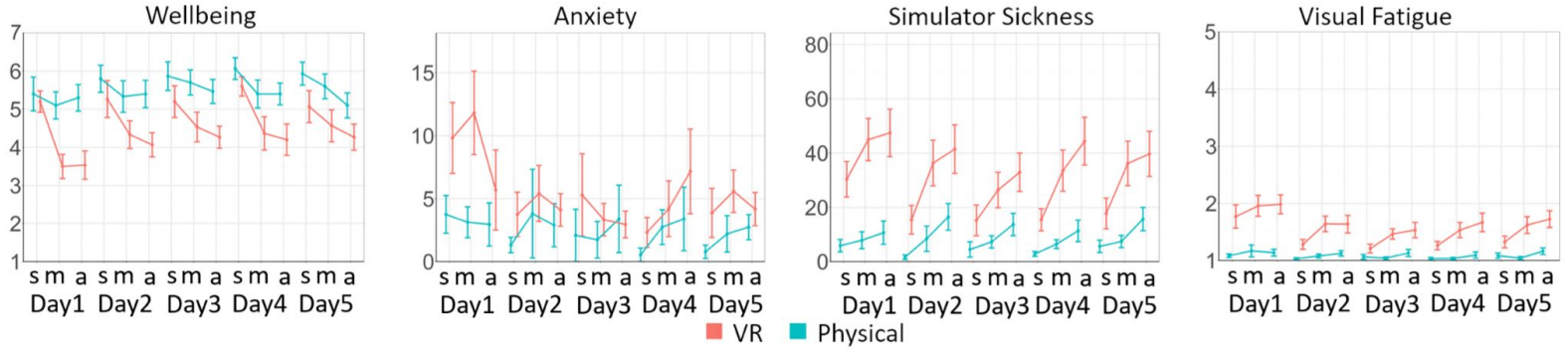
Ruvimova, A., Kim, J., Fritz, T., Hancock, M., & Shepherd, D. C. (2020, April). " Transport Me Away": Fostering Flow in Open Offices through Virtual Reality. In *Proceedings of the 2020 CHI conference on human factors in computing systems*(pp. 1-14).



Unknown long-term therapeutic effects

Answer

Prolonged working in VR and physical shows there are not much significant effects to user actions and movements



Biener, V., Farzinnejad, F., Schuster, R., Tabaei, S., Lindlein, L., Hu, J., ... & Grubert, J. (2023, October). Video Analysis of Behavioral Patterns During Prolonged Work in VR. In *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 466-467). IEEE.
Biener, V., Kalamkar, S., Nouri, N., Ofek, E., Pahud, M., Dudley, J. J., ... & Grubert, J. (2022). Quantifying the effects of working in vr for one week. *IEEE Transactions on Visualization and Computer Graphics*, *28*(11), 3810-3820.

These are early findings but they show some pointers on how to move forward. We call for further studies in the domains of rehabilitation and therapy.

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